BASE CHANGE OF THE CONNECTED COMPONENT SCHEME CUT OUT BY A SECTION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8186-8199 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 8201-8215, the complete author proof (13 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

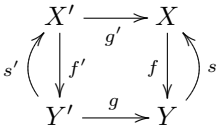
- Explicit prerequisite boundary. The section situation (situation-connected-along-section) defining X^0, the scheme structure on connected components of a fibre, and the geometric connectivity lemma for a connected scheme with a rational point; cited inputs, with the fibre-wise argument retained in full.

Source reference. Original lemma 04PX: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8215–8215. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 055L: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8179–8184. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04KV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8215–8215. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$, $s : Y \to X$ be as in Situation 055L. If $g : Y' \to Y$ is any morphism, consider the base change diagram*



so that we obtain $(X')^0 \subset X'$. Then $(X')^0 = (g')^{-1}(X^0)$.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Let $y' \in Y'$ with image $y \in Y$. We may think of X'_y as a closed subscheme of X_y, see for example Morphisms, Definition 04PX. As $s(y) \in X'_y$ we conclude from Varieties, Lemma 04KV that X'_y is a geometrically connected scheme over $\kappa(y)$. Hence $X'_y \times_y y' \to X'_{y'}$ is a connected closed subscheme which contains $s'(y')$. Thus $X'_y \times_y y' \subset (X'_{y'})^0$. The other inclusion $X'_y \times_y y' \supset (X'_{y'})^0$ is clear as the image of $(X'_{y'})^0$ in X_y is a connected subset of X_y which contains $s(y)$. $\square$